

S. SRINIVASA ARAVINDH

AI Engineer — Applied AI — ML Research

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PROFESSIONAL SUMMARY

AI Engineer with experience building and deploying machine learning and LLM-based applications across healthcare, cybersecurity, document retrieval, and automation domains. Skilled in developing Retrieval-Augmented Generation (RAG) systems, FastAPI-based AI services, model evaluation workflows, and explainable AI pipelines. Experienced in implementing end-to-end ML workflows including dataset preparation, experimentation, model optimization, deployment, and technical documentation for research-oriented and production-focused applications.

EDUCATION

M. Kumarasamy College of Engineering, Karur

Bachelor of Technology in Artificial Intelligence and Machine Learning

Aug 2021 – Jun 2025

CGPA: 8.21

EXPERIENCE

Data Conquest Research Hub

Machine Learning Research Analyst

Coimbatore, India

Mar 2025 – Present

- Architected and deployed scalable, end-to-end machine learning pipelines for complex, high-dimensional datasets in mission-critical domains.
- Developed robust frameworks for model transparency and interpretability, driving trust and actionable insights without compromising on predictive power.
- Engineered advanced data representation strategies and led comprehensive experimental configurations to push the boundaries of model performance.
- Championed reproducible ML research and MLOps-oriented workflows, streamlining transitions from experimental prototypes to publication-ready and deployable AI solutions.

PROJECTS

Production Hybrid RAG System

GitHub Repository

Mar 2026

- Built an async Retrieval-Augmented Generation (RAG) API using FastAPI and LangGraph with hybrid retrieval using FAISS semantic search and BM25 keyword search.
- Improved retrieval relevance through query rewriting, multi-query expansion, and HuggingFace cross-encoder reranking, reducing irrelevant retrieval responses during testing.
- Implemented PDF ingestion, semantic chunking, caching, and SQLite-based metadata storage supporting efficient querying across large document collections.
- Added timing observability and asynchronous request handling, reducing response latency during concurrent document retrieval operations.
- Integrated GitHub Actions CI/CD workflows for automated testing and deployment.

WhatsApp Real-Time E-Commerce & Management System

Production Commercial Deployment

Apr 2026

- Developed a WhatsApp-based commerce platform enabling users to browse products, place orders, and receive order updates through conversational workflows.
- Built chatbot state-management workflows with MongoDB session tracking, cart handling, and automated order processing pipelines.
- Designed a Streamlit-based admin dashboard for live order monitoring and fulfilment management.
- Integrated Twilio WhatsApp APIs and deployed the application using Render with GitHub Actions CI/CD support.
- Improved operational workflow automation for small-scale business order management and customer interaction handling.

MCP Database Assistant — Ollama Mistral + MySQL

GitHub Repository

Feb 2026

- Developed a local AI-powered database assistant capable of converting natural language instructions into MySQL queries using Model Context Protocol (MCP).
- Built MCP client-server communication workflows with FastMCP tool routing and intent parsing mechanisms.
- Integrated multiple LLM backends including Ollama Mistral, Claude, OpenAI, and Gemini within a unified interaction pipeline.
- Designed offline-capable database workflows without dependency on external cloud APIs.

Malware Detection via Dynamic Algorithmic Configuration

IEEE IDCIoT 2025

- Designed a federated learning-based malware detection framework with adaptive algorithm configuration and ensemble optimization.
- Applied explainable AI techniques, Bayesian optimization, and hyperparameter tuning to reduce false positives and improve model reliability.
- Achieved approximately 98% malware detection accuracy on cybersecurity datasets.
- DOI: 10.1109/IDCIOT64235.2025.10914737

Skin Cancer Prediction using Explainable AI — Developed CNN and transfer learning models for dermoscopic image classification with SHAP and LIME-based explainability, achieving 91% classification accuracy.

CAPTCHA Recognition using OpenCV and deep learning techniques.

RESEARCH, PUBLICATIONS & TECHNICAL ENGAGEMENTS

- **Paper:** Malware Detection via Algorithmic Configuration — IEEE Conference, REVA University.
- **Paper:** Skin Cancer Prediction using Explainable AI — National Conference and Journal Publication.
- Speaker at Google DevFest 2023, Madurai on Machine Learning applications and AI workflows.
- Participant at E-Summit 2023, IIT Madras.
- Co-organiser of ARTIFEST 2023 AI Symposium, MKCE.
- Solved 90+ LeetCode problems and completed 728 Skillrack programming challenges.

TECHNICAL SKILLS

LLM & AI Systems: RAG, LangChain, LangGraph, MCP, Vector Databases, Prompt Engineering, Hybrid Retrieval, AI Agents

Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, Transfer Learning, Federated Learning, Explainable AI (SHAP, LIME), Model Evaluation

Backend & APIs: FastAPI, Flask, REST APIs, Async APIs, Streamlit

Databases & Infrastructure: FAISS, BM25, MySQL, MongoDB, SQLite, GitHub Actions, CI/CD, Docker (Basics)

Programming: Python, SQL, C, Java (Basics)

CERTIFICATIONS

Google Cloud Associate Cloud Engineer · AWS Cloud Fundamentals · Oracle Cloud Infrastructure AI Foundations Associate · Machine Learning (ICT Learnathon) · Data Analytics with Python (NPTEL)